The RPLIDAR C1 is a 2D laser scanner used to measure distances around the robot. It provides 360 ° horizontal scanning, with a typical scan frequency of 10 Hz and a maximum specified range of 12 meters on suitable reflective surfaces.

The sensor sends distance and angle measurements to the Raspberry Pi 5 through a UART serial interface. These measurements can be processed to detect obstacles and, when combined with robot localization and a SLAM algorithm, to build a 2D map of the environment.

For GyroBot, the LiDAR is mainly used for environmental perception, obstacle detection and mapping. A separate control system is responsible for maintaining the robot's balance.

RPLIDAR C1

Datasheet gotronic

1. **Laser emission:** the sensor emits a light beam toward an object.
2. **Beam reflection:** part of the light is reflected by the object back to the sensor's receiver.
3. **Distance calculation:** the sensor uses the received light signal to determine the distance, then associates this measurement with a direction.

Distance calculation: Time of Flight (ToF)

The time taken by the light to travel from the sensor to the object and back is measured:

$$d = \frac{c\Delta t}{2}$$

Where:

- **d**: distance between the sensor and the object, in meters;
- **c**: speed of light;
- **Δt**: round-trip time of the light signal;
- the factor 2 comes from the round trip.

Portée maximale annoncée **12 m**	Scan speed **10 Hz** (typical: 10 revolutions/s)
Sampling frequency **5 000 Hz** 5 000 mesures par seconde environ	Précision annoncée **±30 mm** Soit ±3 cm dans les conditions spécifiées

Ce capteur s'alimente en 5 Vcc mais ses entrées et sorties sont uniquement compatibles 3,3 V.

Champ de balayage

360° horizontal, avec une légère étendue angulaire verticale liée à la conception du capteur.

Communication

UART TTL 3,3 V, débit de 460 800 bauds.

Protection

IP54 : protection contre une quantité limitée de poussière et les projections d'eau.

Warning: 12 m of range does not mean 12 m on every object

The range depends notably on the reflectivity of the surface:

Surface type	Specified range
White surface, 70% reflectivity	Up to 12 m
Black surface, 10% reflectivity	Up to 6 m

A dark surface generally absorbs more light and returns less signal to the receiver. Lighting conditions, object orientation and geometry can also affect measurement quality.

4. Angular resolution

The typical specified angular resolution is 0.72°. This is the angular gap between successive samples, not a guarantee that every detail of that size will be detected.

For a full revolution (5,000 measurements/s ÷ 10 revolutions/s = 500 points per revolution):

$$N \approx \frac{360}{0{,}72} = 500$$

This gives **500** angular positions per revolution at this resolution.

5. Data processing chain

RPLIDAR C1 (distance and angle measurements) → Onboard computer: Raspberry Pi 5 (acquisition and processing of measurements) → Mapping algorithm (associates scans with the robot's position) → 2D map of the environment (walls, obstacles and free areas)

SLAM

For mapping, the algorithm used will be SLAM, which stands for Simultaneous Localization and Mapping. It consists of two simultaneous tasks:

1. Estimating where the robot is.
2. Building or updating the map of the environment.

connect the RPLIDAR C1 to the Raspberry Pi 5

Communication interface: UART

Le C1 utilise une liaison série UART TTL à 3,3 V, avec un débit nominal de 460 800 bauds et un format série 8N1, liaison vu en cours.

The C1 uses a 3.3 V UART TTL serial link, with a nominal baud rate of 460,800 and an 8N1 serial format (as seen in class).

The two main signals are:

- TX (Transmit): data transmission.
- RX (Receive): data reception.

A common GND is also required. Wiring must follow the C1 datasheet and the connector used.

Software

Several software tools exist, such as:

- Slamtec software
- ROS 2
- Nav2
- RViz2

To compare: which of these are actually able to perform SLAM.

For comparison

Coordinates and position

The LiDAR naturally provides a distance **r** and an angle **θ**. To build a map, it is therefore important to convert these correctly into Cartesian coordinates:

donc :

- $x = r\cos(\theta)$
- $y = r\sin(\theta)$

Example: the sensor detects a wall at 2 meters in the 30° direction.

$$x = 2\cos(30°) \approx 1,73\ m \mid y = 2\sin(30°) = 1\ m$$

The point is therefore located at about (1.73; 1) meters from the sensor, in the reference frame used for this example.